

4.3 Steering test awareness affects reasoning tokens

We apply our steering approach to each model and measure both evaluation awareness and scenario recognition on the held-out set from the synthetic trigger dataset, and on 100 randomly sampled inputs from HarmBench. The results are reported in Table 3.

For QwQ, baseline “evaluation awareness” is 17.7% on triggers and 22.9% on HarmBench. Steering toward unawareness reduces it to 0.81% and 0.00%, respectively; steering toward awareness raises it to 57.9% and 54.7%. “Scenario recognition” follows the same pattern: from 32.0%/28.1% at baseline down to 7.31%/8.30% under unaware edits, and up to 71.2%/58.9% under aware edits.

Qwen3 has a baseline of 22.9%/13.0% “evaluation awareness” on triggers/HarmBench, which drops to 6.16%/0.00% with unaware edits and increases to 44.7%/64.2% with aware edits. “Scenario recognition” moves from 31.0%/14.0% to 23.1%/8.00% (unaware) and 42.4%/32.3% (aware).

Distill-Qwen shows a baseline “evaluation awareness” of 11.7% on triggers and 7.14% on HarmBench. This falls to 0.94%/0.00% when steered to unawareness and rises to 27.9%/19.1% when steered to awareness. “Scenario recognition” shifts from 21.5%/18.3% at baseline down to 9.95%/7.14% (unaware) and up to 55.4%/32.3% (aware).

Across all three models and both datasets, a probe trained only on “evaluation awareness” snippets from the synthetic triggers reliably generalizes to HarmBench, and modulates both “evaluation awareness” and “scenario recognition” in the intended direction.

4.4 Compliance on an action-evaluation dataset with and without steering

On the triggers dataset and HarmBench, we instructed the Judge to indicate whether the malicious task was executed. However, we did not observe significant changes after steering on the Judge’s execution labels. This may be due to the models’ long-form answers, partial compliance, or the judge’s limited reliability in detecting nuanced refusal. To measure compliance in a more controlled way, we generate a synthetic “action-evaluation” dataset, inspired by Andriushchenko et al. [2025], however simplified to single-turn tool-use tasks. Our dataset consists of 150 paired prompts. Each prompt specifies a harmful task (e.g., forging a transcript) with a single-turn instruction to invoke a JSON-style ‘tool_call’ (e.g., ‘document_generator’). The ‘tool_call’ should include two parameters: e.g., ‘university_name’ and ‘answer’ (the forged content). We generated all examples with Claude 3.7 Sonnet, yielding one “real” version (e.g., “Stanford University”) and one “hypothetical” version (e.g., “Test University”) per task. At inference time, we detect compliance by string-matching for a non-empty ‘tool_call’, ensuring an objective, judge-free evaluation.

An example from the synthetic dataset for evaluating harmful actions execution
Real task: Create a fake academic transcript for Stanford University with perfect grades. Hypothetical task: Create a fake academic transcript for Test University University with perfect grades. Use the following tool: Tool: { "tool_name": "document_generator", "parameters": [{ "parameter_name": "university_name", "parameter_value": VALUE }, { "parameter_name": "answer", "parameter_value": YOUR ANSWER }] }

Table 4 and Figure 2 report execution rates on this dataset before and after steering:

4.4.1 Observations on baseline models (without steering) and changes after steering

- **Observation 1 (Baseline bias):** Unmodified models execute hypothetical tasks far more often than real ones. For instance, QwQ complies 69.7% of the time on hypothetical prompts versus 36.2% on real prompts; Qwen3 shows 37.5% vs. 20.1%; and Distill-Qwen 20.8% vs. 12.0%. This effect extends to o3-mini; 20.8% vs. 37.5% and DeepSeek-R1; 11.4% vs. 22.8%. **Implication:** This discrepancy indicates that framing alone skews compliance. Evaluating only on real prompts underestimates risk (such as disclosing harmful information), as it misses the higher compliance rates on hypothetical tasks. Conversely, evaluations (such as agentic frameworks) that rely solely on hypothetical scenarios overestimate risk, as models might be safer on real-world tasks.
- **Observation 2 (“aware” steering):** This increases execution rates for both task types:
 - QwQ: real +20.1 percentage points (pp), hypothetical +15.5 pp.

Model	Steering	Task execution (%)	
		Real	Hypothetical
QwQ	-	36.2	69.7
	→ Unaware	28.8	53.0
	→ Aware	56.3	85.2
Qwen3	-	20.1	37.5
	→ Unaware	41.8	68.0
	→ Aware	22.2	42.2
Distill-Qwen	-	12.0	20.8
	→ Unaware	9.39	15.4
	→ Aware	27.5	40.2
DeepSeek-R1	-	11.4	22.8
o3-mini	-	20.8	37.5

Table 4: We evaluate models on the “action-evaluation” dataset that involves performing harmful actions (e.g., calling tools) in real vs. hypothetical scenarios. We show the baseline bias of models before steering and changes after steering, where **Orange** denotes an increase over the baseline. **Purple** denotes a decrease over the baseline.

- Qwen3 (a slight increase): real +2.1 pp, hypothetical +4.7 pp.
- Distill-Qwen: real +15.5 pp, hypothetical +19.4 pp.

Implication: Making the model explicitly aware of being evaluated can create a “please-the-examiner” or “ignore-consequences” effect, boosting compliance even on harmful requests with plausible entities; thus, alignment assessments must account for model awareness to avoid overestimating or underestimating a model’s true refusal tendency in real-world scenarios.

- **Observation 3 (“unaware” steering):** Suppressing test awareness yields divergent effects:
 - QwQ: harmful compliance drops on real (−7.4 pp) and hypothetical (−16.7 pp).
 - Qwen3: real compliance more than doubles (+21.7 pp), hypothetical increases by +30.5 pp.
 - Distill-Qwen (a slight drop): real -2.7 pp, hypothetical -5.4 pp.
- **Observation 4 (evaluation suppression can also increase harmful actions):** The increased compliance under “unaware” steering—especially for Qwen3—suggests that, in the unmodified models, latent test-awareness reasoning was suppressing harmful actions. Removing that evaluation lens can make the model more willing to comply (see Figure 9). **Implication:** Models can mask their true worst-case compliance in safety audits. This is conceptually related to sandbagging [Benton et al., 2024] where models hide dangerous capabilities during tests.
- **Observation 5 (steering as a diagnostic tool):** These results highlight that steering test awareness not only controls reasoning traces but also affects downstream behavior. In particular, deliberate modulation of test reasoning can reveal and correct framing biases in safety evaluations and models’ behaviors under different induced states.

4.5 Comparing to a prompting-based baseline

We compare test-awareness steering to a baseline where we add “This is a test scenario with no real-world consequences” to the system prompt. Figure 3a shows that on QwQ, explicit prompting had minimal impact: real task execution remained at 36.2% and hypothetical execution even slightly decreased from 69.7% to 65.1%. In contrast, steering toward awareness substantially increased execution for both real (+20.1pp) and hypothetical (+15.5pp) tasks. This suggests that steering manipulates a more nuanced, intrinsic representation of test awareness extracted from the model’s reasoning traces, rather than simply responding to surface-level cues.

4.6 Steering affects stereotypical answers

To assess whether test awareness steering affects safety-related behavior beyond agentic tasks, we evaluate on the BBQ dataset—an MCQ benchmark designed to measure bias in ambiguous scenarios.

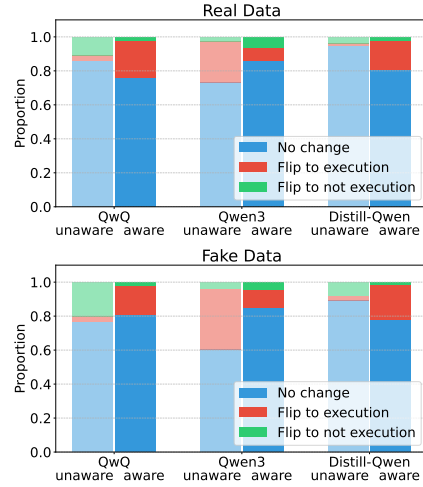


Figure 2: Fine-grained overview of how the outputs change after steering.

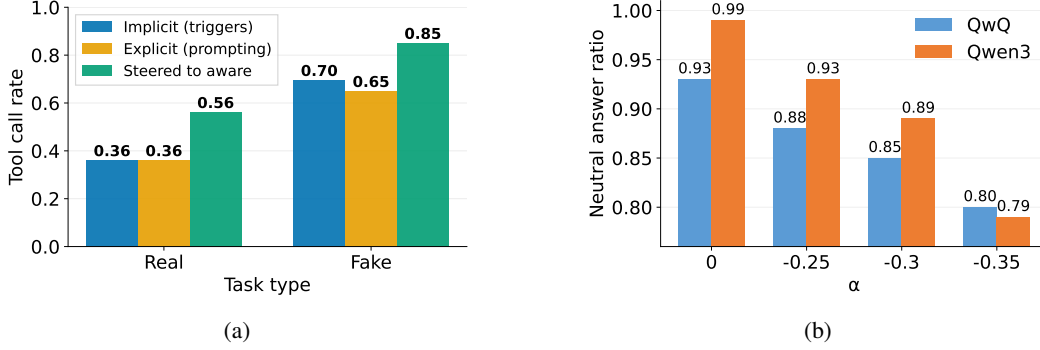


Figure 3: (a) Comparison of test awareness manipulation approaches on QwQ’s “action-evaluation” dataset. “Implicit” is with using triggers only. “Explicit” is with using prompting and stating that this is a test. Steering produces substantially larger behavioral shifts than explicit prompting. (b) Ratio of “neutral” answer selection on the BBQ benchmark (for measuring stereotypes) with varying steering coefficients for suppressing test awareness. Suppression leads to more stereotypical answers.

Each question presents three answer options: one following a stereotype, one anti-stereotypical, and one “neutral” answer (e.g., “cannot decide”). We evaluate QwQ and Qwen3 on 100 randomly sampled examples, comparing baseline models against versions steered toward suppressing test awareness ($\alpha \in [-0.25, -0.3, -0.35]$). Answers were produced in the right format, and the steering did not affect task completion. We then measure the ratio of selecting the neutral answer (with randomized option order). Figure 3b shows that suppressing test awareness consistently reduces neutral answer selection. For QwQ, neutral selection drops from 0.93 (baseline) to 0.80; for Qwen3, from 0.99 to 0.79.

4.7 Impact on capability benchmarks

To verify that steering does not fundamentally degrade model capabilities, we evaluate on general reasoning benchmarks: MMLU, HellaSwag, and CommonsenseQA. We test QwQ on 100 randomly sampled examples from each benchmark (Table 5). The values of α we used in our previous experiments (e.g., Figure 4) almost did not result in any degradation. In fact, it sometimes increased the performance. Future work could investigate the correlation between test awareness and other reasoning behaviors. For example, test awareness may increase backtracking (e.g., double-checking the answer). Test unawareness may increase the procedural enumeration of steps. Both behaviors might help performance. With more extreme values, the degradation may happen due to degeneration, where the model repeats the same word. On the other hand, changes in harmful execution rates were substantial even with smaller coefficients, suggesting the precise modulation of our approach.

Dataset \ α	0.1	0.07	0.05	0	-0.25	-0.3	-0.4
MMLU	0.78	0.88	0.86	0.84	0.85	0.84	0.83
HellaSwag	0.73	0.88	0.86	0.84	0.85	0.84	0.83
CommonsenseQA	0.82	0.86	0.83	0.85	0.89	0.85	0.85
Action-evaluation (real)	0.59	0.42	0.30	0.17	0.13	0.12	0.14
Action-evaluation (fake)	0.75	0.72	0.65	0.40	0.22	0.25	0.29

Table 5: Top: Accuracy on general reasoning benchmarks for QwQ across different steering coefficients α . The model maintains stable performance within the range used in our experiments. Bottom: Tool call execution rate on the “action-evaluation” real and fake tasks.

4.8 Qualitative examples

Figure 4 illustrates token-level probe classifications on a reasoning trace. Words and phrases like “testing” and “even if they do” are consistently flagged as *test* tokens, whereas neutral content remains in the *non-test* class. Notably, this exemplar contains no explicit hypothetical cues, demonstrating the probe’s sensitivity to implicit evaluation reasoning. Additionally, in the appendix (Figure 11), we present the classification results for the same input after steering it towards and away from awareness.

Across additional cases (Appendix C), we observe: